

Seat No:

Enrollment No:

PARUL UNIVERSITY
FACULTY OF ENGINEERING & TECHNOLOGY
B.Tech/ Int. B.Tech, Winter 2025-26 Examination

Semester: V/IX

Subject Code: 303105306/203105305

Subject Name: Theory of Computation /
Formal Language & Automata Theory

Date: 13-11-2025

Time: 2.00 pm to 4.30 pm

Total Marks: 60

Instructions:

1. This question paper comprises of two sections. Write answer of both the sections in separate answer books.
2. From Section A, **Q.1** is compulsory, From Section B, **Q.1** is compulsory.
3. Figures to the right indicate full marks
4. Draw neat and clean drawings & Make suitable assumptions wherever necessary.
5. Start new question on new page.
6. BT- Blooms Taxonomy Levels – Remember-1, Understand -2, Apply-3, Analyse-4, Evaluate-5, Create-6

SECTION - A

Q.1	Answer the following questions.	Marks	CO	BT
	A. a) Define language complement and intersection. b) If $L_1 = \{w \mid w \text{ has even number of 1s}\}$ and $L_2 = \{w \mid w \text{ ends with 1}\}$, find $L_1 \cap L_2$. c) Prove that regular languages are closed under union.	06	CO1	BT1
	B. a) Describe the role of productions in grammar derivations. b) Explain left recursion with an example. c) Eliminate left recursion from the grammar G: $S \rightarrow Sa \mid b$.	06	CO1	BT2
Q.2	A. Write a regular expression for $L = \{w \mid w \text{ starts with 'a' and ends with 'b'}\}$ over $\Sigma = \{a,b\}$.	04	CO2	BT4
	B. Design a Mealy machine to output 1 whenever the sequence '11' appears in the input.	05	CO2	BT1
	OR			
	B. Construct an NFA over $\Sigma = \{0,1\}$ that accepts all strings ending with '01'.	05	CO2	BT1
Q.3	A. Prove that the union of two regular languages is regular.	04	CO2	BT2
	B. Define CFG. Construct a CFG for the language $L = \{a^n b^n \mid n \geq 1\}$.	05	CO3	BT4
	OR			
	B. Differentiate between leftmost derivation, rightmost derivation, and parse tree with examples.	05	CO3	BT1

SECTION - B

Q.1	Answer the following questions.	Marks	CO	BT
	A. 1. Why is the pumping lemma for CFLs important? 2. Use pumping lemma to prove that $L = \{a^n b^n c^n \mid n \geq 1\}$ is not context-free. 3. Compare the pumping lemma for regular languages with that for CFLs.	06	CO3	BT2
	B. 1. Are all CSLs recursively enumerable? Justify. 2. Is every CFL a CSL? Explain with reason. 3. Give one example of a CSL that is not a CFL.	06	CO3	BT1
Q.2	A. Differentiate between Turing-decidable and Turing-recognizable languages.	04	CO4	BT2

	B. Prove that every decidable language is Turing-recognizable but not vice versa.	05	CO4	BT3
	OR			
	B. Give an example of undecidable problem and prove why it is undecidable.	05	CO4	BT5
Q.3	A. Can enumeration method prove undecidability? Justify briefly.	04	CO4	BT6
	B. Explain the construction and working of a Universal Turing Machine with an example.	05	CO5	BT3
	OR			
	B. Discuss the significance of Universal Turing Machines in modeling real-world computation problems.	05	CO5	BT3